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PALEOMAGNETISM OF THE SAMAIL OPHIOLITE, OMAN
1. THE SHEETED DIKE COMPLEX AT IBRA

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Abstract. We have studied the rock magnetism and paleomagnetism of 200 oriented samples from the middle Cretaceous sheeted dike sequence of the Samail ophiolite, Oman. The samples are from 28 sites spanning a cross-strike distance of 30 km. Most of the dikes have undergone greenschist metamorphism. The natural remanent magnetization (NRM) of the samples is about 1 A/m and Q_n is also near 1. Thermal and alternating field demagnetization treatment resolves both normal (north) and reversed (south) polarities. About one third of the samples have low median destructive fields (MDF) less than 100 Oe (7.96×10^3 A turn/m). These low MDF samples have scattered directions and are classified as undetermined polarities (U). Hysteresis measurements show pseudo-single domain carriers predominate in all samples, although the low MDF samples are suggested to have the largest grains in the samples studied. Thermomagnetic curves show both magnetite and minor maghemite. Thermal demagnetization and microscopic study reveals that hematite is abundant and is the carrier of the R remanence. In addition, several dikes exhibit both N and R polarities. These facts, coupled with the degree of metamorphism, indicate that part of the remanence is secondary and could be due to hydrothermal metamorphism below the seafloor. We suggest that the N polarity is primary and resides in exsolved magnetite, while the R polarity is secondary and resides primarily in metamorphic-origin hematite. Several short R events observed in the Albian through Cenomanian magnetostratigraphic record are candidates for the R event in the ophiolite if the R polarity originated near the spreading center. Alternately, nonreproducible self-reversal is another possible origin for the R polarity. Regardless of whether the reversal originated at the spreading center, secondary magnetization in the dikes considerably increases the directional dispersion. Another major factor affecting dispersion is viscous remanent magnetization (VRM) in the larger magnetite grains. This extreme dispersion disallows the Samail ophiolite dike layer as a significant source for marine magnetic anomalies, regardless of the strength of the NRM or the value of Q.

The stable N and R directions yield a paleomagnetic pole which is far sided, suggesting northward movement of the ophiolite. However, the paleomagnetic pole is not as far sided as the contemporaneous Africa pole, which suggests a component of north-south closure between the ophiolite and Africa prior to northward movement.

Introduction

In trying to deduce the source of marine magnetic anomalies due to seafloor spreading and magnetic field reversals, several questions are important: (1) What types of oceanic rocks have a net vector natural remanent magnetization (NRM) which is both strong and stable enough to contribute significantly to anomalies, (2) when was the magnetization acquired, (3) what is the nature of magnetic properties of these rocks, and (4) what is the geometry of the magnetization contrasts (boundaries) which give rise to the anomalies? To date, many studies have concentrated on the basalts of oceanic layer 2 as a source candidate (e.g., see review of Johnson [1979]. The deeper rocks of the oceanic crust have received less attention, mainly because they are less well exposed. However, the excellent exposures often available in ophiolites allow their magnetism to be studied and, under certain circumstances, to be used as models for oceanic crust magnetization [Levi et al., 1978; Banerjee, 1980; Swift and Johnson, 1982]. We are studying the paleomagnetism of the sheeted dike sequence of the Samail ophiolite, Oman, in order to address the above questions. Not all of them are answered or discussed here. In the following text we report our findings on the main magnetic properties of the sheeted dike unit and present the first observation in an ophiolite body of what may be a magnetic polarity reversal recorded during seafloor spreading. In a companion paper [Luyendyk and Day, this issue] we discuss the results from a study of the Samail gabbroic and ultramafic rocks near the village of Ibra.

The Samail Ophiolite

The ophiolite occurs as a west to northwest trending massif along the Gulf of Oman (Figure 1). A thorough reconnaissance study of the ophiolite was reported by Glennie et al. [1974]. It is mainly Cretaceous in age and was obducted in the Late Cretaceous soon after its formation [Coleman, 1981]. Along with other investigators from the U.S. Geological Survey, the California Institute of Technology, the University of Texas, the University of Hawaii, and the University of California, Santa Barbara, we studied a portion of the ophiolite in a region near the villages of Ibra and Thabiti (Figures 1 and 2).

In the study area the ophiolite occurs as a nappe which dips gently (~15°) to the south [Pallister, 1981; Bailey, 1981]. The dike sequence is exposed in a broad open valley which is either an east-west trending asymmetric syncline, almost vertical on the south limb, or a

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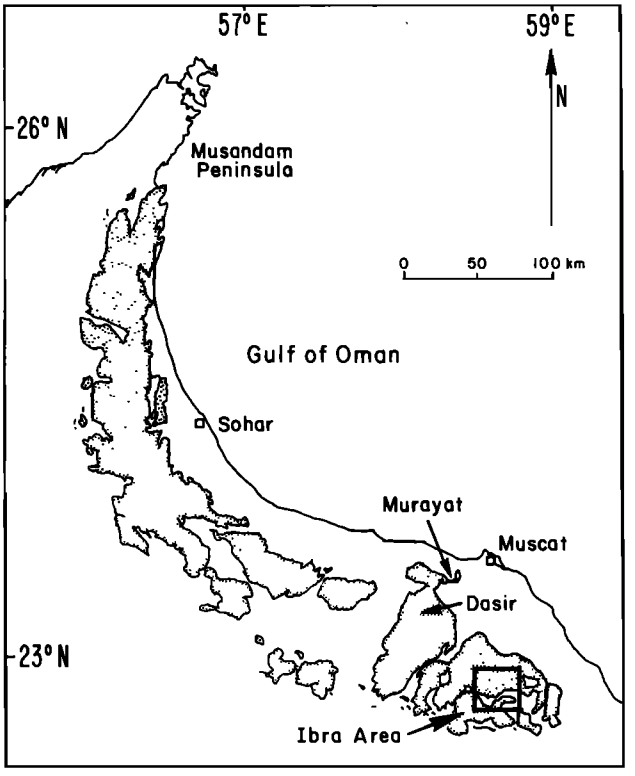


Fig. 1. Map of the Sultanate of Oman region showing the Oman Mountains, Samail ophiolite, and the study area.

gentle homocline reverse faulted at its south end (Figure 2). The sheeted dikes in the valley strike approximately north and have a stratigraphic thickness of 1.2 to 1.6 km [Pallister, 1981]. A minor amount of capping extrusives are in contact with the top of the dike sequence near the center of the study area, while in the west the lower dike-gabbro contact can be seen [Bailey, 1981] (Figure 2). The entire ophiolite section here is over 6 km thick [Hopson et al., 1981]. Radiogenic U-Pb age determinations are between 93.5 and 97.9 m.y. in this region [Tilton et al., 1981]. The strike of the dikes plus the geometry of the cumulate gabbro layers suggest that the isochrons trend north-northwest here and that the ancient ridge axis lays to the east. The age distributions imply that the spreading occurred at a very fast rate [Tilton et al., 1981].

The petrology of the dike rocks is discussed by Pallister [1981]. The dikes are diabase which has undergone hydrothermal metamorphism at the spreading center and subsequent retrograde metamorphism. They exhibit actinolite-calcic plagioclase, greenschist, and lower metamorphic facies. Most of the pyroxene has been replaced by uralite, presumably at temperatures above 550°C [Liou et al., 1974], probably at the spreading center. Evidence from oxygen isotope studies confirms that hydrothermal circulation thoroughly penetrated the sheeted dikes [Gregory and Taylor, 1981]. The metamorphic alteration varies at each outcrop and often within a single dike.

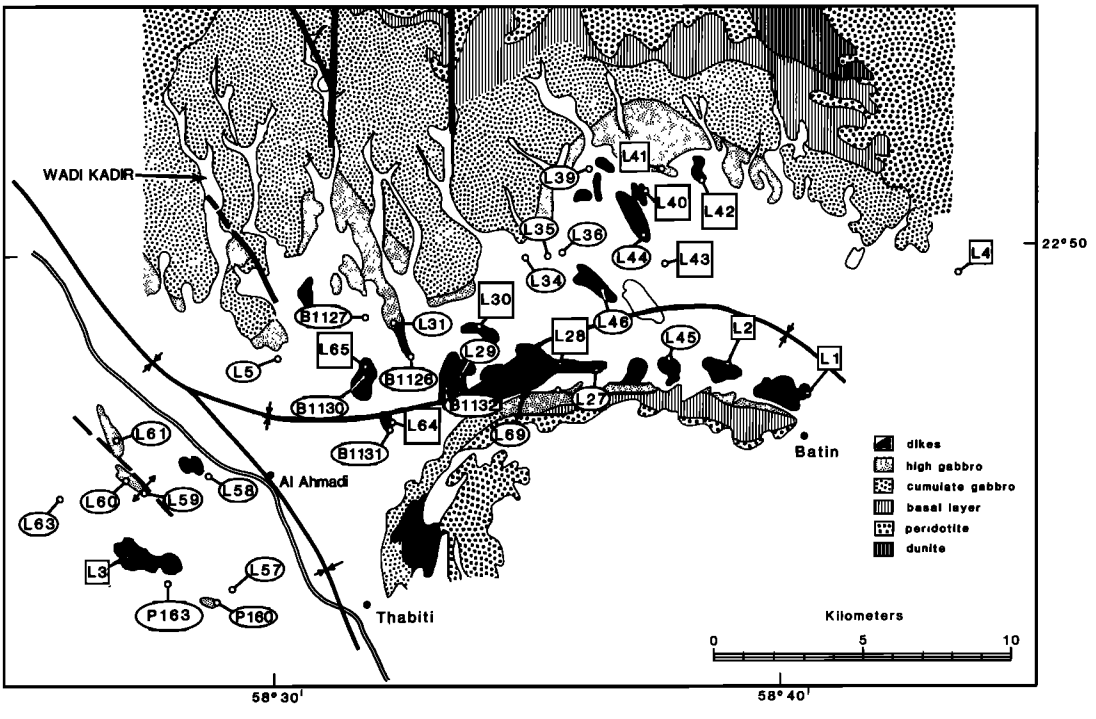


Fig. 2. Simplified geologic map of the ophiolite nappe near Ibra and Thabiti, after Bailey [1981] and Pallister [1981]. Dike outcrops are shown in solid black, and sampling sites are labeled. Site numbers in boxes are ones on which rock magnetic studies were made.

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TABLE 1. Summary of Magnetic Properties of Dike Rocks

Variable	Number of Observations	Mean	Standard Deviation	Median	Geometric Mean
NRM, A/M	202	2.99	9.46	0.726	0.603
MDF, Oe	166	206	172	155	150
Susceptibility, SI	185	5.35 E-2	0.363	2.16 E-2	1.82 E-2
Q_n*	180	4.31	17.3	0.791	0.823
H_C, Oe	35	239	106	225	213
H_{RC}, Oe	44	522	176	488	493
J_R, A/m	35	689	340	610	615
J_S, A/m	30	3180	2300	2460	2460

*For a field of 0.5 T.

Recently, a controversy has arisen concerning whether this ophiolite originated at a mid-ocean ridge. Although the predominance of workers favor this origin, Pearce et al., [1981] have presented evidence which they believe favors an origin in a back arc basin. C. Hopson (unpublished manuscript, 1982) has countered this argument on several points. The origin of the ophiolite, although clearly pertinent to our purpose, is beyond the scope of this paper. At present, we are favorably inclined toward Hopson's points in favor of an origin at a mid-ocean ridge.

Sampling

Over two hundred 2.5-cm-diameter oriented samples were taken from 40 sites. At most sites the dikes were sampled with a portable gas-powered drill and oriented with a Brunton-type compass. However, at many locations the dikes were too brecciated to drill, and oriented hand samples were taken which were later cored in the laboratory. In some cases, two sister samples (2.5 x 2.5 cm) could be obtained from the drilled core. Our site locations span over 30 km in the cross-strike direction (Figure 2). A very detailed sampling program was carried out at site L3 which corresponds to Pallister's [1981] Al Ahmadi section.

Laboratory Measurements

NRM, χ, Q_n

The natural remanent magnetization (NRM) of the samples has a median value near 0.73 A/m (10^{-3} emu/cm^3), an arithmetic mean near 2.99 A/m and a geometric mean of 0.6 A/m (Table 1). However, quite a number of samples are either weaker than 10^{-1} A/m or stronger than 10 A/m (Figure 3). The geometric mean value is close to that found for Macquarie Island dikes [Levi and Banerjee, 1977] but greater than that of a Chilean ophiolite [Stern et al., 1976], the Point Sal ophiolite [Kempner, 1977], the Othris dike rocks [Beske-Diehl and Banerjee, 1979], and the Bay of Islands dikes [Swift and Johnson, 1982], and less than the value for the Smartville diabase rocks [Banerjee, 1980]. A value of 1.3 A/m was found for the Troodos massif sheeted dikes [Vine and Moores, 1972]. The arithmetic average value we find is somewhat high compared to other reported values and is in fact more comparable to averages usually found for the extrusive layer in ophiolites [Banerjee, 1980]. However, our average value is biased by a few very strongly magnetized samples.

The geometric mean values of the low field susceptibility and the Koenigsberger ratio Q_n (calculated for a field of 0.5 T) are similar to those found for other ophiolite bodies (Table 1). The Q_n value of near 1 is lower than the average value of 4 or 5 we found for the Samail gabbros [Luyendyk and Day, this issue].

Thermomagnetic Curves

Thirty samples from 12 sites were selected for rock magnetic studies (Table 2). Thirteen of these samples are from the Al Ahmadi site (L3). Not enough core sample was available to duplicate the rock magnetic work on cores which also had both alternating field demagnetization (AFDM) and thermal demagnetization (TDM) treatments (discussed below).

Thermomagnetic measurements were made on magnetic separates of 30 samples. The curves were run at 5 x 10^3 Oe (3.98 x 10^5 A turn/m) at a vacuum of 10^{-2} torr. Both reversible and irreversible heating-cooling curves of J_S-T were observed (Figure 4).

All but one sample showed Curie points of magnetite at 580°C; one sample gave a higher temperature suggesting hematite. Microscopy observations suggest that magnetite has exsolved from titanomagnetite in these samples. This probably occurred during high-temperature oxidation below the seafloor [Butler et al., 1976; Petersen et al., 1979; Kono, 1980; Banerjee, 1980].

The unmixing of titanomagnetite into magnetite and ilmenite has been noted in greenschist facies pillow basalts from the Troodos massif [Levi and Banerjee, 1977], the Macquarie Island ophiolite [Butler et al., 1976], and the Smartville ophiolite [Levi and Banerjee, 1977]. Additionally, dike samples from these ophiolites and the northern part of the Oman ophiolite have also given magnetite-like Curie points [Levi and Banerjee, 1977; Banerjee and Moscowitz, 1979].

Reversible thermomagnetic curves were found for 14 samples, and irreversible curves were found for 16 samples. Four sites have samples with both types of curves. Reversible or irreversible curves are seen for samples from the remaining eight sites (Table 2).

As is seen in Figure 4, the irreversible curves show a loss of saturation magnetization near

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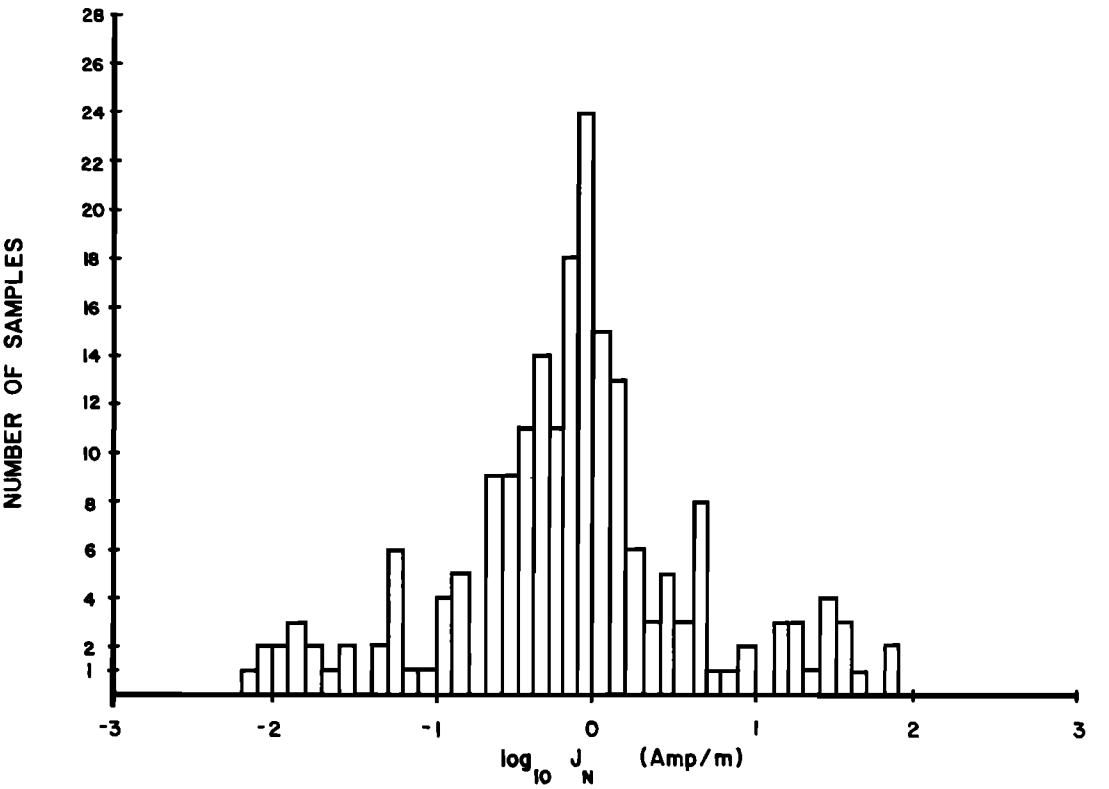


Fig. 3. Histogram of NRM values found for the dike rocks. One A/m equals 10^{-3} emu/cm^3.

TABLE 2. Rock Magnetic Properties of 30 Samples

Sample	J_N, A/m	χ, SI[a]	Q_n	MDF, Oe	Polarity[b]	J_S-T[c]	J_R, A/m	J_S, A/m	H_{RC}, Oe	H_C, Oe
L 1.04	2.24	2.25	0.80	90	R	Ir	219	880	450	169
L 1.08	1.59	7.55	0.53	95	U	Ir	368	3540	190	42
L 2.03	1.03	2.12	1.22	225	N	Re	633	2860	660	197
L 3.01	0.204	2.69	0.19	75	U	Re	377	1890	380	126
L 3.03	0.106	9.69	0.03	210	U	Re	618	1950	730	338
L 3.04	1.28	1.02	3.14	O.S.[d]	R	Ir	471	2150	615	281
L 3.05	0.028	0.82	0.08	O.S.	R	Ir	435	1280	850	380
L 3.06	0.199	0.95	0.53	160	N	Ir	573	1290	930	478
L 3.07	0.373	0.83	1.13	250	N	Ir	322	960	607	225
L 3.08	0.747	0.84	2.22	485	N	Ir	525	1300	830	422
L 3.09	0.322	0.66	1.22	370	N	Re	304	875	700	338
L 3.10	0.043	0.60	0.18	O.S.	R	Ir	298	773	666	281
L 3.13	9.69	0.33	73.7	40	U	Ir	141	328	870	338
L 3.18	0.605	0.12	1.26	370	N	Ir	673	1640	790	352
L 3.21	0.410	1.19	0.86	400	R	Re	530	2770	615	281
L 3.23	0.132	1.37	0.24	430	U	Re	447	3320	520	211
L 4.02	1.60	6.03	0.67	155	N	Re	901	5550	350	113
L 4.12	0.372	3.94	0.24	370	R	Re	1000	3900	430	169
L 28.03	0.143	3.57	0.10	50	U	Ir	661	3270	320	127
L 30.01	0.967	1.88	1.29	225	N	Ir	592	2460	440	169
L 30.04	0.838	5.59	0.38	45	U	Re	535	3590	320	84
L 40.05	26.4	1.03	64.5	125	R	Ir	579	1560	770	338
L 41.03	0.330	5.09	0.16	85	N	Re	984	4780	413	141
L 41.07	0.636	1.51	1.06	65	U	Re	587	2100	675	239
L 43.05	25.6	3.33	19.3	40	R	Ir	219	880	130	56
L 64.04	1.11	3.89	0.72	165	N	Re	4020	748	430	127
L 65.04	0.783	1.03	1.91	925	R	Re	1220	5550	490	169

——— in sample list separate cooling units.

[a] SI units x 10^{-2}.

[b] See text for definition.

[c] Re, reversible and Ir, irreversible thermomagnetic curves.

[d] Off scale or no MDF by 1000 Oe.

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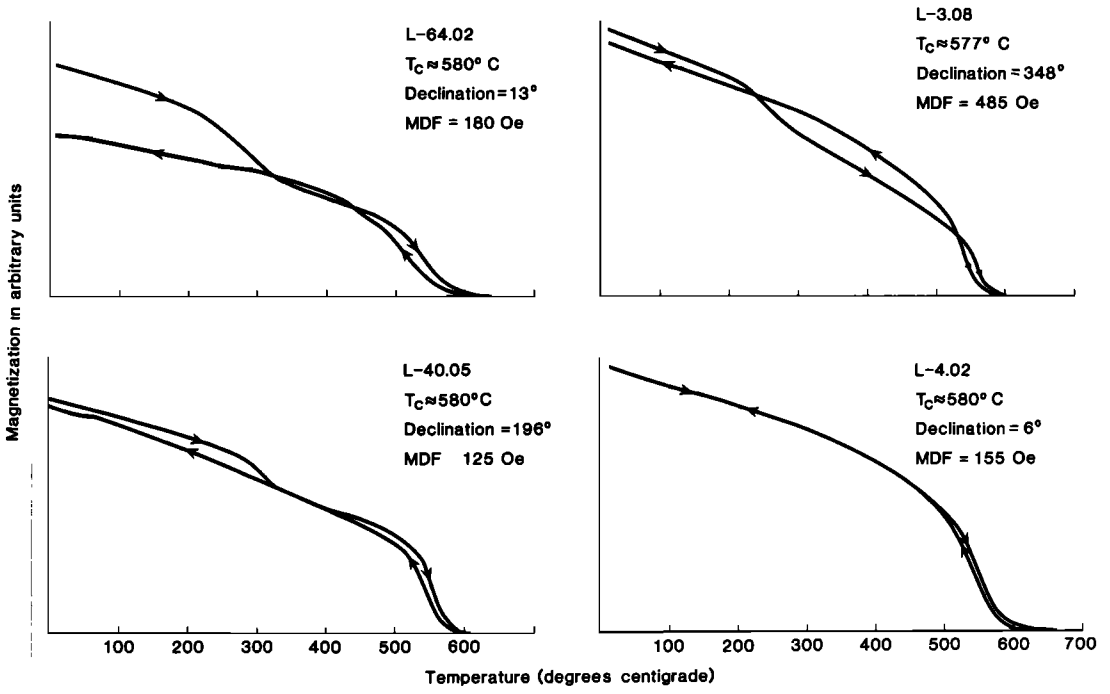


Fig. 4. Four representative thermomagnetic curves. Samples 64.02, 3.08, and 40.05 contain varying amounts of maghemite, which is indicated by a saturation magnetization drop near 300°C.

300°C and an irreversible magnetization loss on cooling. This thermomagnetic behavior can be attributed to the conversion of thermally unstable maghemite to hematite during heating [Nagata, 1961]. The magnetization drop was strongly exhibited in only four samples.

Maghemite was also detected in the Troodos ophiolite [Levi and Banerjee, 1977], the Othris ophiolite [Beske-Diehl and Banerjee, 1979], and in the northern part of the Oman ophiolite [Banerjee and Moscowitz, 1979]. Banerjee [1980] has argued that since maghemite has yet to be reported in submarine rock samples, it probably represents a subaerial oxidation product. Beske-Diehl and Banerjee [1979] also found that samples with maghemite possessed a hard viscous remanent magnetization (VRM), which was believed to have been acquired after ophiolite obduction.

Reflected Light Microscopy

Polished thin sections of 30 samples were studied under reflected light using oil immersion lenses (Table 3). Most of the samples are the same ones appearing in Table 2. The percentage of opaque minerals in each slide is commonly between 1 and 5%, although a maximum of 7-10% was observed in one slide. The size range of visible crystals is 5 to 200 μ across but crystals 40 to 80 μ seem to predominate. The opaque minerals are magnetite, hematite, and small amounts of ilmenite (Table 3). Oxidation of magnetite to maghemite is suspected, but maghemite could not be distinguished from hematite due to the predominance of the latter and the fine-grained nature of the replacement textures. The relative proportion of magnetite and hematite is variable. Hematite dominates 14 slides, magnetite dominates nine, and on seven slides the percentages are roughly the same.

Most opaque crystals are subhedral to euhedral grains which are often skeletal and usually have several well-preserved crystal faces. These grains are magnetite with varying percentages of hematite. The hematite in these crystals pseudomorphically replaces the magnetite. In crystals of low hematite content the hematite invades the edges of the crystal along crystallographic planes. In slides showing greater magnetite oxidation the magnetite grain may be completely replaced with hematite. This magnetite-to-hematite reaction could have occurred during deuteric alteration, retrograde metamorphism, late hydrothermal alteration, or subaerial oxidation. Ilmenite exsolution is found as internal and external composite inclusions and as sandwich lamellae. These same inclusions and lamellae are visibly replaced by leucoxene, chlorite, and sphene and/or calcite.

Hematite is also present in the form of finely divided anhedral to subhedral grains measurably less than 20 μ. It is commonly amorphous and displays bright red internal reflections. Yellow internal reflections suggestive of geothite are also common. These fine hematite grains are disseminated in secondary minerals and are probably products of hydrothermal activity. Iron oxides in this form compose a major fraction of the opaques in only two samples.

Hysteresis (see below) and thermomagnetic measurements indicate that a predominant magnetic remanence carrier is PSD magnetite. Since PSD magnetite is in the size range of ~1 to 10 or 20 μ [Day et al., 1977], we expect that many of these grains were undetected with the microscope. Even though the relative proportion of hematite

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TABLE 3. Mineralogy of Polished Thin Sections Determined Using Petrographic and Reflecting Microscopes

Sample	Polarity	Percent Opaques	Mt	Il Grains	Il Incl	Ht From Mt	Ht	Ca	Sp	Qt	Ze	Ep	Act	Chl	Oxy Class	T Block
L3.01	U	1-2	20	-	5	75	Tr	X	X	-	?	-	X	X		
L3.02	N	1	70	-	5	25	Tr	X	?	-	-	-	X	X	1	
L3.03b	U	1-2	40	-	Tr	55	5	?	X	-	-	st		X	3	
L3.05	R	3-5	10	-	Tr	85	5	X	X	-	?	X	-	X		
L3.06	N	1-2	5	-	-	90	5	X	X	-	?	?	X	X	1	
L3.06b	N	3	5	-	Tr	90	5	X	X	-	-	-	X	X		
L3.07	N	3	15	-	-	82	3	X	X	-	?	?st	r	X	3	
L3.08	N	2	25	-	Tr	75	-	X	X	-	?	?	r	X	1	
L3.09	N	1-2	5	Tr	Tr	90	5	X	X	-	?	X	r	X	4+6	
L3.10	R	1	35	-	Tr	60	5	X	X	-	?	?	r	X	4+5	
L3.12	U	1-2	50	15	5	30	Tr	?	X	-	-	-	X	X	3	
L3.15	R	1	40	-	-	60	Tr	X	X	-	-	-	X	?	3	
L3.16	N	1	35	2	3	60	Tr	?	-	-	-	?	X	X	3	
L3.18	N	1-3	25	7	5	63	Tr	X	X	-	-	X	X	X	3+4	
L3.19	R	1-3	1	5	9	85	5	X	X	-	-	?	?	X	4	700
L3.21	R	2-3	30	10	10	50	Tr	X	?	-	?	-	X	X	3+4	
L3.22	R	1-2	85	-	5	10	Tr	X	X	-	-	?	X	X	3+4	700
L3.23	U	2	40	-	20	40	-	X	?	-	X	X	X	X		
L3.25	N	3-4	45	-	2	45	8	X	-	-	?	-	X	X		
L3.26	R	2-3	70	-	-	30	Tr	X	?	-	-	X	X	X		700
L4.02	N	3-5	50	20	25	5	Tr	X	-	X	-	-	X	X	2	
L4.12	R	5	80	-	3	15	Tr	X	-	X	-	X	X	X	3	
L28.03	U	?	-	-	-	20	80	X	-	X	-	-	X	X	1	
L30.01	N	3	60	-	15	15	-	X	-	X	-	-	X	X		
L30.04	U	3	90	-	8	2	-	X	-	-	-	-	X	X	3	
L41.03	N	7-10	10	-	-	90	-	X	-	-	-	-	X	X	2+3	
L41.07	U	3-5	60	-	-	40	Tr	X	-	X	-	-	X	X	2+3	
L43.05	R	1	-	-	-	20?	80	X	-	X	-	-	-	-	2	
L64.04	N	1-2	80	5	-	15	Tr	X	-	X	-	-	-	X	2+3	610
L65.04	R	5-7	23	-	13	60	5	X	-	X	-	-	-	X	2	

Mt, magnetite in percent of total opaques. Il grains, ilmenite occurring as discrete grains, given in percent of total opaques. Il incl, ilmenite as composite inclusions and lamellae in magnetite given in percent of total opaques. Ht from Mt, hematite from oxidation of magnetite, given in percent of total opaques. Ht, hematite as internally reflective grains disseminated in transparent minerals (some may be lepidocrosite and/or geothite), given in percent of total opaques. Ca, calcite; Sp, sphene; Qt, quartz; Ze, zeolite; Ep, epidote; Act, actinolite; Chl, chlorite. Crosses indicate presence, question marks indicate possible presence, and dashes indicate absence. r, remnant actinolite in chlorite; st, stilpnomelane in epidote. All percentages are approximate.

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observed is often considerable, its saturation magnetization is, of course, much less than that of magnetite. Thus, the absence of hematite-like Curie temperatures (i.e., very high temperatures) in the thermomagnetic analyses is not surprising (Figure 4). Similarly, only minor amounts of maghemite were observed in thin section. This tends to be in agreement with the J_S-T curves, which mostly show only a slight drop in J_S upon heating. The few samples which display maghemite in thin section also have irreversible loss of J_S in thermomagnetic experiments; however, maghemite was not visually observed in all samples that show irreversible curves.

The slides exhibit a mineral suite consistent with a complex metamorphic history. Several distinctive alteration minerals were noted. They include actinolite, probably a deuteric alteration product of pyroxene and/or hornblende; chlorite, after actinolite indicative of greenschist metamorphism (retrograde); calcite and quartz which can result from deuteric or retrograde alteration of plagioclase but are most probably of hydrothermal origin; epidote, a product of plagioclase sausseritization (deuteric) or retrograde metamorphism; sphene, a greenschist facies alteration product, its Ti probably comes from ilmenite destruction; and zeolite (?), the result of very low grade retrograde metamorphism.

The oxidation state observed for the opaque minerals was compared with the classification used by Ade-Hall et al. [1971], who studied opaque mineral response to hydrothermal alteration. Characteristics of both deuteric and regional hydrothermal metamorphism are apparent. Most common are grains with class 2 deuteric oxidation without hydrothermal alteration effects and class 3 deuteric oxidation along with a wide range of hydrothermal alteration. Of all the samples viewed, over 75% show hydrothermal effects on the opaque minerals. The most extreme oxidation is a class 3 deuteric alteration with magnetite replaced by hematite and ilmenite replaced by sphene.

Several of the grains show significant departures from the classification scheme of Ade-Hall et al. [1971]. These grains exhibit textures similar to examples of class 3 deuteric oxidation, but the details of the mineralogy differ significantly. In the Ade-Hall et al. examples of class 3 deuterically altered grains which have not been hydrothermally altered, the grains are crisscrossed by a trellis pattern of ilmenite-hematite solid solution lamellae. The lamellae appear lighter in color than the host magnetite, presumably due to a significant fraction of Fe_2O_3 in solution in the lamellae. The host magnetite in some of our samples, however, contains trellis lamellae of ilmenite which are darker brown than the host magnetite, suggesting a low hematite fraction in the lamellae. In these same grains there are also trellis lamellae of a phase lighter than the host magnetite. This lighter phase could be maghemite, hematite, or ilmenite-hematite solid solution with a high enough hematite mole fraction to render it lighter colored than the magnetite.

We have not been able to resolve the paragenesis of the lamellae by textural relations. It seems probable, however, that the dark ilmenite lamellae represent familiar high-temperature ilmenite exsolution lamellae and that the lighter lamellae are a later, possibly lower-temperature oxidation phase.

We could not find any exsolution textures within the ilmenites or ilmenohematites. Ilmenite alteration was limited to pseudomorphic replacement by darker nonreflective phases or by development of 'woven textures' of rutile, Ti-hematite and pseudobrookite [Ade-Hall et al., 1971, figure 7]. According to Ade-Hall et al. both of these alterations occur at low-temperature burial metamorphism conditions.

Both of the above ilmenite alteration phenomena can yield hematite; the latter 'woven texture' contains Ti-hematite and the replacement of ilmenite lamellae (e.g., $2FeTiO_3 + \frac{1}{2}O_2 \rightleftharpoons 2TiO_2 + Fe_2O_3$) yields Fe available to form secondary hematite, which may be represented by the red internally reflective grains found disseminated in all of the samples.

Demagnetization Studies

Alternating field demagnetization (AFDM) series were run on 202 samples. The AFDM of the samples shows a wide range of characteristics (Figures 5 and 6). About 28% of the samples have

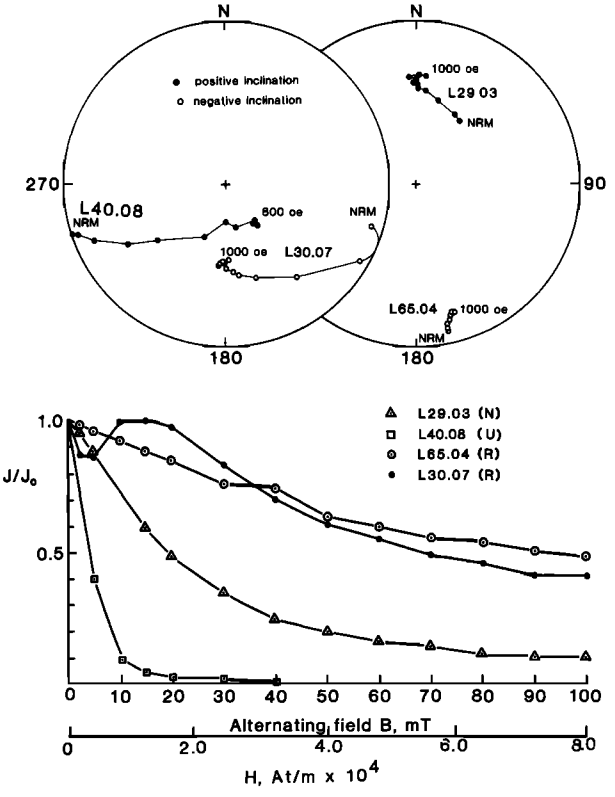


Fig. 5. Some representative AF demagnetization curves for the dike samples. N, normal polarity; R, reversed polarity; U, undetermined polarity. Also shown are stereographic projections of vector direction changes upon demagnetization for the same samples. Sample L40.08 has the lowest MDF. The geocentric axial dipole direction is at I = 40, D = 0, and the present field direction is I = 33, D = 0.

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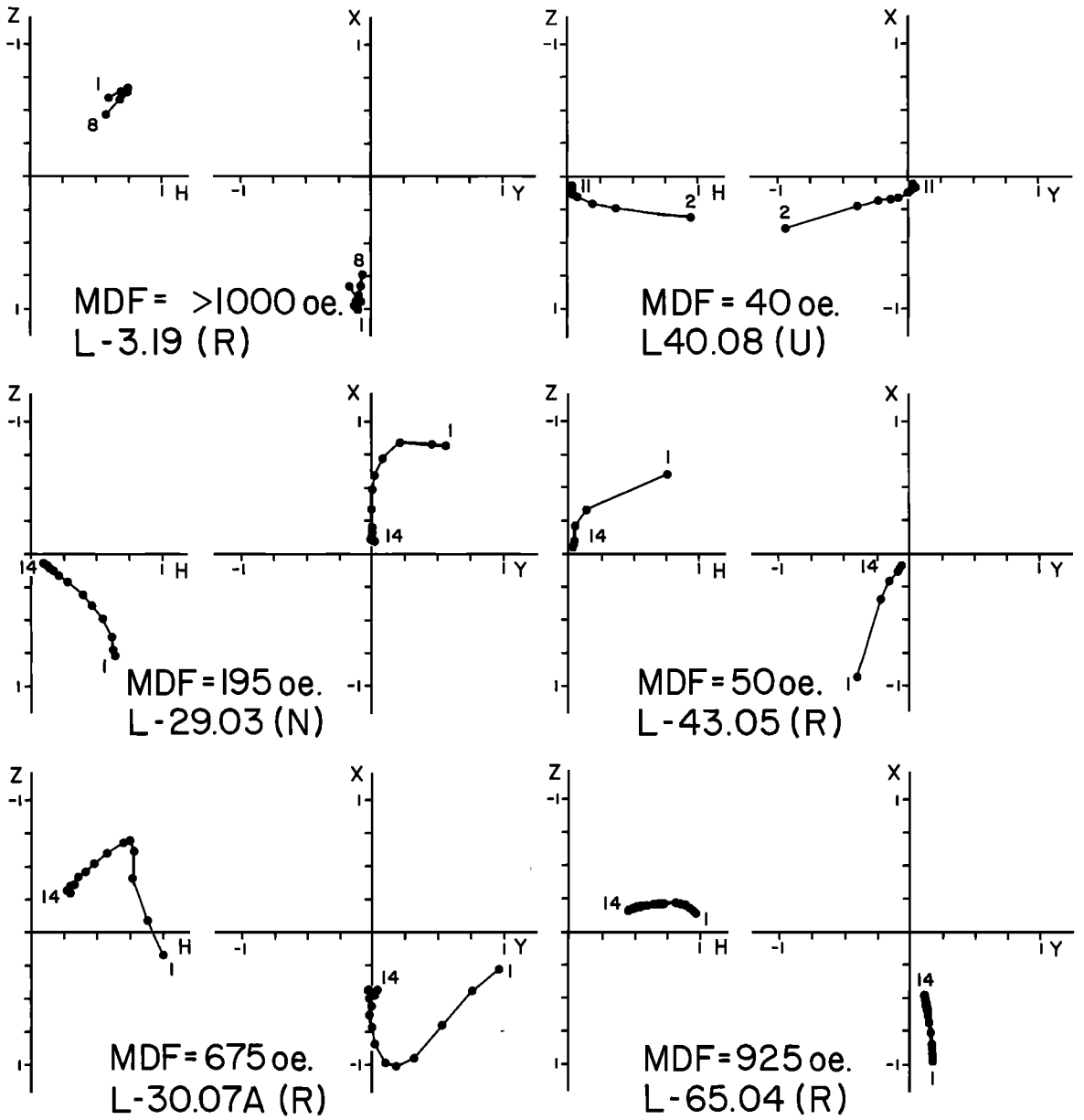


Fig. 6. Orthogonal vector component diagrams of progressive demagnetization series from 0 to 1000 Oe (0 to 7.96 x 10^4 A turn/m). The Z-H diagram shows inclination behavior, and the X-Y diagram shows declination (X is north).

a median destructive field (MDF) less than 100 Oe (7.96 x 10^3 A turn/m). The remainder show demagnetizing curves which are stable to AFDM with MDF's ranging to over 1000 Oe (7.96 x 10^4 A turn/m; Table 1). The softer samples are either directionally unstable or their stable directions are randomly distributed (discussed later). Of the total samples, 99 are classed as normally magnetized, 45 as reversed, and 58 as of undetermined (or unstable) polarity. The stable direction could usually be determined at demagnetization levels between 300 and 500 Oe, regardless of polarity.

We selected the normal polarity samples (N) as those with stable declination within ±40° north, the reversed samples (R) as those with stable declinations with ±° of south, and the undetermined polarities (U) as those with no stable direction or stable directions other than N or R. During demagnetization, about 20% of the samples showed clear multiple directions of magnetization (e.g., Figure 6). About 75% of these (n = 30) can be interpreted as having a softer N component along with a harder R or U component. Incomplete removal of the N component results in more directional scatter for the samples classified as R compared to those classed as N.

The arithmetic average MDF for both normal and reversely magnetized samples is around 230 Oe (1.83 x 10^4 A turn/m, Table 5). However, about 14% of the reversed samples and only a few normal samples have an MDF greater than 1000 Oe. Inspection of a cumulative frequency distribution of MDF for N, R, and U samples shows that N and R

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samples have similar distributions in the lower ranges of MDF but that there are significantly more R samples with MDF higher than 300 or 400 Oe compared to N or U samples (Figure 7). Although a low coercivity normal component on the reversed samples biases their MDF to higher values, we believe the correct interpretation of Figure 7 is that the R magnetization is generally harder than the N or U magnetizations.

Thermal demagnetization (TDM) was performed on 19 samples. Samples were heated in a low field (<100 gammas) (nT) at increments of 50°C up to 650°C. The TDM directional behavior in all but three samples closely followed the AFDM directional behavior of sister samples from the same core (Figures 8 and 9).

Seven of the 19 samples displayed more than 15% of their NRM intensity remaining at 600°C. This residual moment disappeared after heating to 700°C. This result suggests the presence of hematite in some samples, which is responsible for a significant portion of the remanence. Optical microscopy observations confirm this suggestion. In addition, the samples with high blocking temperatures are all of R or U polarity, and the ones with lower blocking temperature are N (Figure 8). Thus the AFDM and TDM experiments both suggest that the R (and U?) polarity is due to a chemical remanent magnetization (CRM) in hematite. This CRM was either acquired subaerial on the outcrop or during hydrothermal alteration beneath the spreading center [Petersen et al., 1979]. However, the intense degree of high-temperature oxidation effects seen in the samples, plus the fair degree of directional coherence of many of the R samples, suggests that a subseafloor high-temperature CRM is the most likely origin for the R remanence.

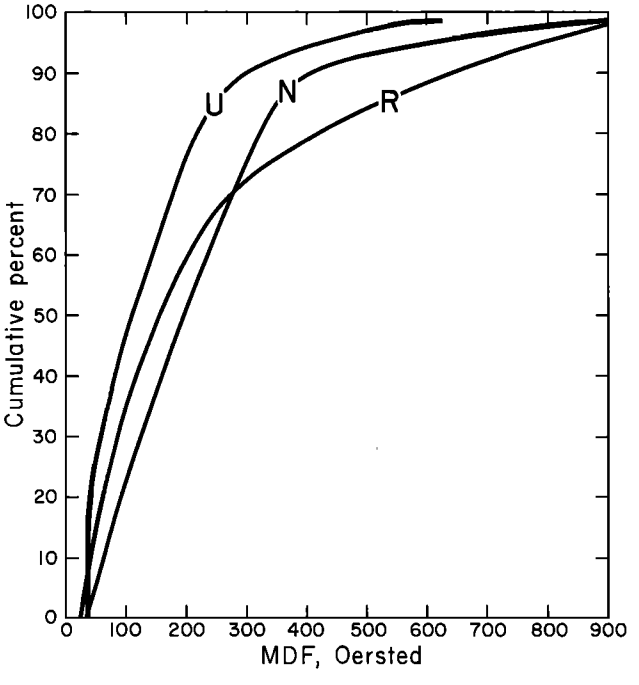


Fig. 7. Cumulative frequency curves for the MDF of all samples classed according to polarity. The R sample set has more samples with very high MDF than do the N or U sets.

The TDM experiments also suggest that some of the remanence may be carried by maghemite. An intensity drop near 300°C was observed for eight samples (see Figure 8, right). This may be due to conversion of maghemite to hematite on heating. The directional data suggest that the maghemite (?) is of N polarity.

Hysteresis Measurements

Hysteresis loops were measured on 30 samples with a vibrating sample magnetometer capable of 10×10^3 Oe (7.96×10^5 A turn/m) magnetizing fields (Table 2 and Figure 10). Each loop was corrected for paramagnetic contributions and read for values of coercive force H_C, saturation remanence J_R, and saturation magnetization J_S. Remanent coercive force H_{RC} was determined by DC demagnetization of isothermal remanence acquired at 5.7×10^3 Oe.

The ratios J_R/J_S and H_{RC}/H_C are plotted against each other in Figure 11 in order to estimate grain size of magnetic carriers [Day et al., 1977]. The samples studied appear to contain the full range of pseudo-single domain (PSD) sized magnetic particles. The samples with lower MDF (<100 Oe) mostly have larger grains according to this analysis (Figure 11).

Viscous Magnetization Acquisition

VRM was induced in four samples from three sites. The purpose of the experiment was to see if significant amounts of VRM could be acquired in a short time and how stable or hard any acquired VRM might be.

The samples were first demagnetized at 1000 Oe (7.96×10^4 A turn/m) and then placed in a solenoid with a 0.5-Oe uniaxial field for 1500 hours. Remanence was measured at five subsequent times.

The samples tested included magnetically hard (high MDF of NRM) normal and reversed samples and soft normal and reversed samples. The samples with high MDF of NRM did not acquire a detectable VRM, but the soft ones increased their remanence by 2 times during the experiment (Figure 12). These samples which acquired VRM were then AF demagnetized. The MDF of VRM was not different from the MDF of NRM.

Beske-Diehl and Banerjee [1979] detected a hard VRM which was rapidly acquired in samples from the Othris ophiolite. This hard VRM showed directions close to the present field. They suggested that it was due to the presence of maghemite and/or magnetite.

Although our VRM experiment is not exhaustive, it did not detect hard VRM. In addition, the demagnetization behavior of samples with irreversible J_S-T curves (suggesting maghemite) is not clearly different from the behavior of those samples suspected of having only magnetite.

AFDM Characteristics of NRM Versus TRM

Progressive AFDM was conducted on seven samples from five sites which had been imparted a total TRM. The TRM was acquired in a 0.75-Oe field at a temperature near 675°C. In the TRM sets there were specimens of normal polarity,

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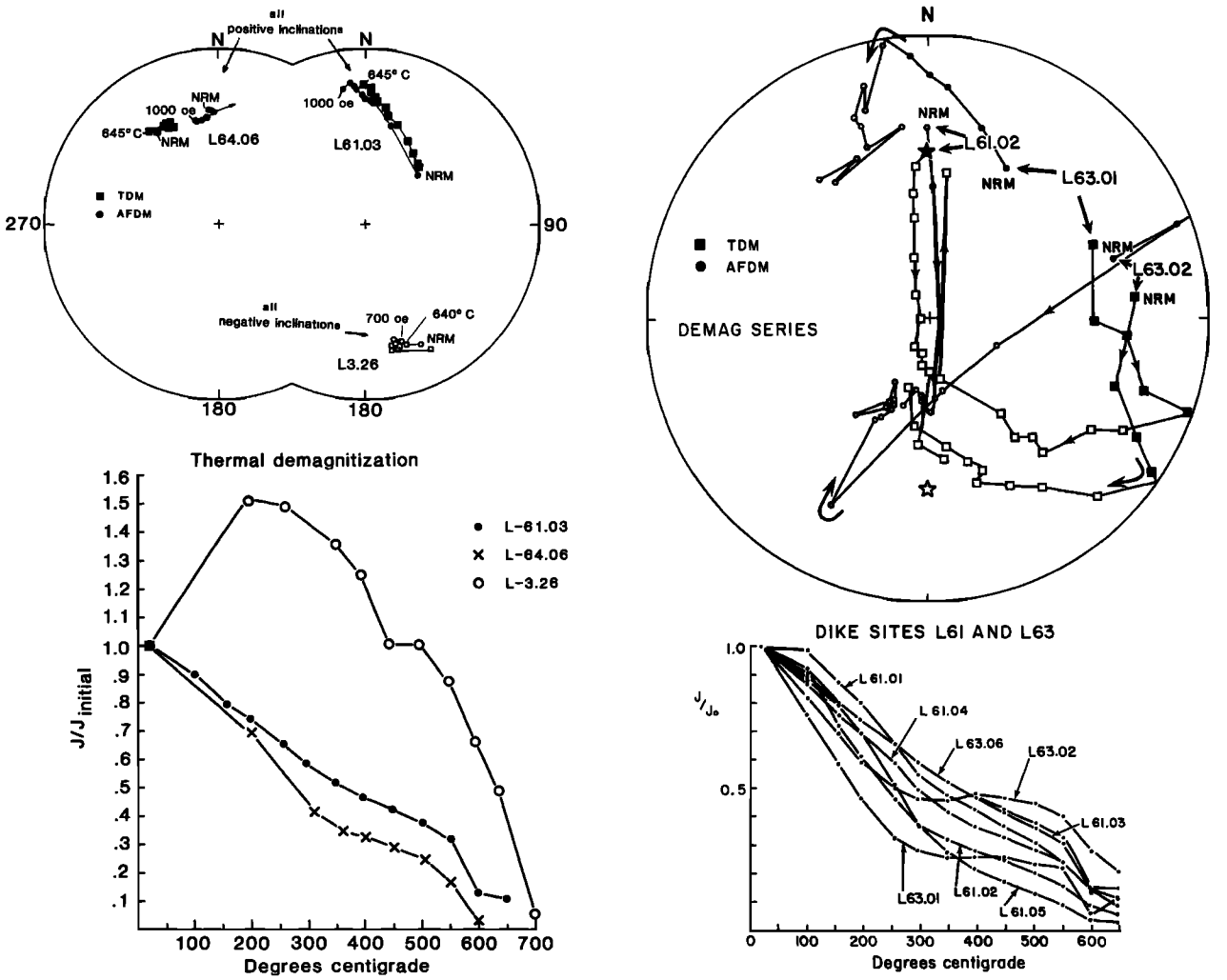


Fig. 8. (lower) Normalized intensity plots of thermal demagnetization series and (upper) stereogram presentation of the demagnetization behavior of the TDM samples compared to behavior of sister samples during AFDM. Solid star is dipole field direction and open star is the reverse. Sample 3.26 shows 30% hematite and 70% magnetite in polished section.

reversed polarity, MDF of NRM less than 100 Oe and MDF of NRM greater than 100 Oe. Table 4 presents a comparison of MDF values of NRM and TRM.

For samples with an MDF or NRM over 100 Oe, there is good correlation between the MDF obtained from the AFDM of NRM and that of TRM. For two samples with an MDF of NRM less than 100 Oe the MDF of TRM is significantly greater than that of the NRM. For these samples (30.05 and 43.05) the stable direction is obtained after removing 87 and 99% of their NRM intensity, respectively. Clearly, most of the NRM in these low MDF samples is VRM, as might be expected. Two of the high MDF of NRM samples from site L64 do have MDF values for NRM and TRM differing by up to 70 Oe. This may indicate that even though a sample carries a predominantly TRM-like remanence, it may still be capable of acquiring a small viscous component.

An inherent problem in an experiment such as this is the possibility of chemical alteration due to heating, leading to erroneous analogies or conclusions. To check this, hysteresis loops were obtained from 17 samples from eight sites both before and after heating to nearly 675°C. The shape of the before and after loops remained reasonably similar.

Discussion of Experimental Results

The rock magnetic parameters we measured allow us to make the following statements:

1. The NRM, χ, and Q_n values for the dikes are slightly higher than values reported for other ophiolites but are most similar to the values for the Troodos ophiolite. Our values are also comparable to those reported for sill rocks sampled on DSDP leg 30 [Dunlop and Prevot, 1982].

2. Thermomagnetic curves and reflected light observations show the presence of exsolved magnetite. This probably is due to deuteric oxidation of titanomagnetite in the ocean crust. Hematite is present and could be due to hydrothermal alteration. Irreversible loss of J_S during thermomagnetic experiments suggests the presence of maghemite in about half the samples tested.

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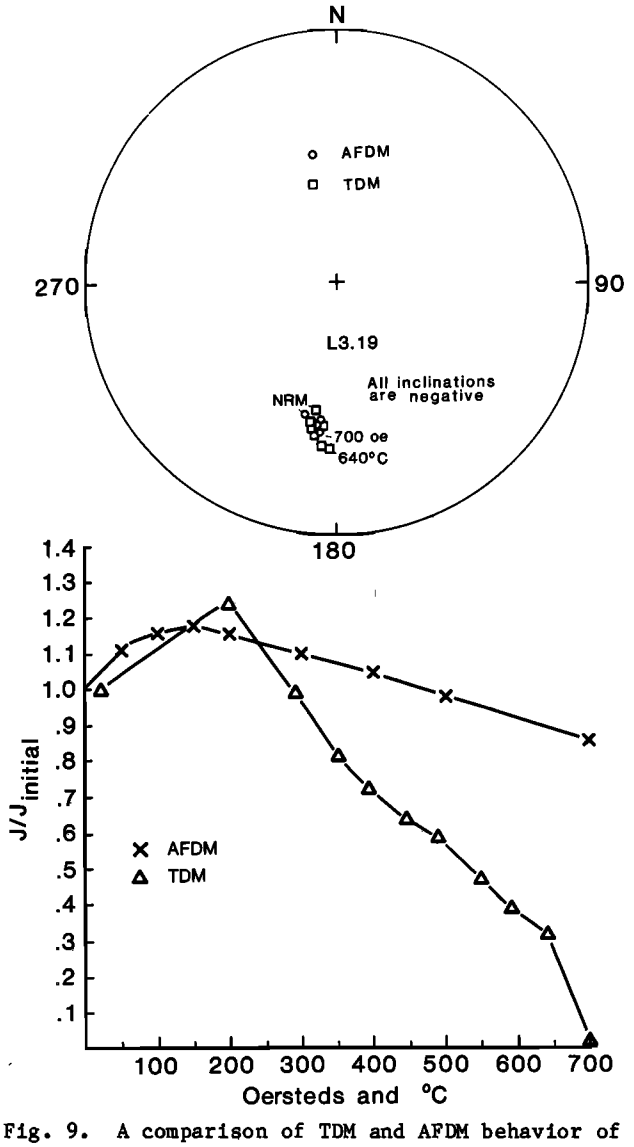


Fig. 9. A comparison of TDM and AFDM behavior of sample L3.19 showing unusually high coercive force and blocking temperatures probably indicative of fine-grained hematite. Microscopic investigation of this sample shows that the magnetic mineral is entirely hematite.

3. Normal polarity, reversed polarity, and undetermined polarities (intermediate directions) were observed. The intermediate directions are largely but not exclusively associated with samples having an MDF <100 Oe (7.96×10^3 A turn/m). Table 5 shows that the MDF of samples with undetermined polarity (U) is significantly lower (95-100% level) than the MDF of N and R samples.

4. The samples with low MDF (and mostly U polarity) appear to have larger grain size of magnetic carriers (nearer to multidomain) than samples with high MDF. This is indicated by hysteresis measurements (Figure 11) and a significantly higher H_{RC} for N and R samples compared to U samples (Table 5) [see Day et al., 1977].

5. The behavior of sister samples during thermal and AF demagnetization is very similar in almost all cases. TDM behavior suggests that hematite carries a significant portion of the remanence in some samples. This remanence is of R polarity. Magnetite and maghemite carry N polarity. TDM behavior also suggests that some U polarities are due to combination of N and R directions and not only VRM.

6. The N or R polarities are not correlated with any measured magnetic property other than blocking temperature. This is evident in the 't' test by polarity (Table 5). Also, we performed a contingency analysis, which did not detect that the NRM, MDF, and susceptibility are contingent on whether the rock is polarized N, R, or U.

7. Demagnetization of TRM acquired in the laboratory suggests that the NRM of high MDF samples is probably TRM or CRM and the NRM of low MDF samples is viscous remanence.

The statistical interdependency of rock magnetic properties was studied. A correlation coefficient table was constructed for NRM, susceptibility, Q_n, MDF, H_C, H_{RC}, J_R, J_S, and J_R/J_S. In addition, multiple regression analysis was applied to NRM, susceptibility, and MDF. Susceptibility is strongly correlated with J_S ($r = 0.47$ at 99%). Analysis of variance shows that 35% of the variation in susceptibility is related to variation in J_S ($F = 14.4$ at 99.9%). Kono [1980] also noted this result in rocks from the Hawaiian-Emperor seamount chain and concluded that suceptibility was controlled by the concentration of magnetic material. NRM is directly correlated with susceptibility ($r = 0.51$ at 99.9%) and J_S ($r = 0.47$ at 99%) and inversely related to MDF ($r = 0.15$ at 95%). For NRM, 36% of its variation is related to susceptibility and MDF^{-1} ($F = 44.4$ at 99.9%). The MDF is directly correlated with H_{RC} ($r = 0.36$ at 97%). A multiple regression analysis for MDF showed that 42% of its variation is inversely related to susceptibility, Q_n, and J_S and directly related to J_R ($F = 3.76$ at 97%). These results suggest that susceptibility and NRM are directly related to increases in the concentration of magnetic grains. MDF is inversely related to both magnetic grain size and concentration. This points to a model also used by Kono [1980], whereby slow cooling of magmas favors an increase in both the concentration and grain size of magnetic minerals.

Paleomagnetic Directions

Field Directions

Each core we drilled provided at least one sample which was subjected to progressive AFDM to find a stable paleomagnetic direction. The stable directions were determined mostly by inspecting stereoplots of demagnetization series for directions which changed the least on incremental cleaning. These selected directions were then cross-checked in several ways. In some cases, independent selection of stable directions was made using the paleomagnetic stability index of Symons and Stupavsky [1974]; in others, orthogonal vector diagrams were used (Figure 6). For some samples, the vector difference was calculated for demagnetization series and the stable direction picked when the total vector direction and vector difference direction were the same. As was expected, all three cross-checking methods

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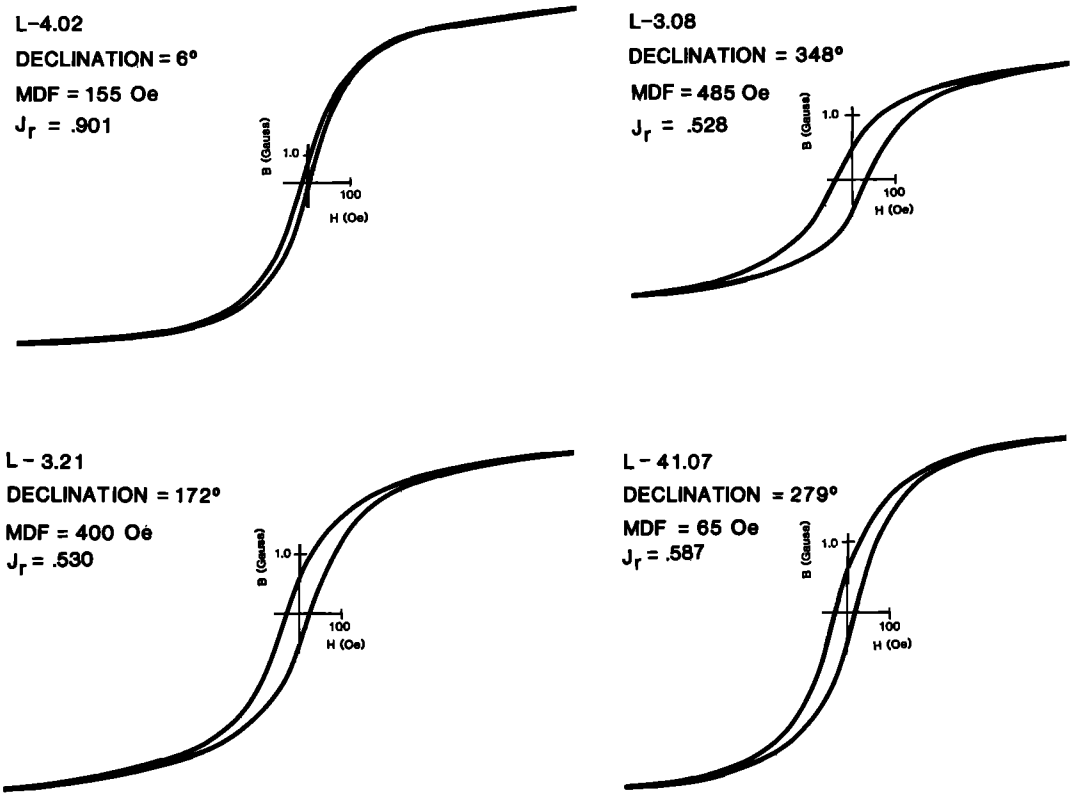


Fig. 10. Four representative hysteresis loops illustrating contrasts in behavior.

yielded the same stable direction as that obtained from stereonet inspection.

Compared to the NRM directions, the selected stable directions show slightly improved clustering (Figures 13a-13c). Most notably, the stable selections show a clustering of reversed directions, indicating the removal of a normal component on demagnetization. The NRM directions show a very definite cluster on the present earth's field direction (Figure 13a). In Figure 13c, we have further selected the data by eliminating directions for samples with MDF <100 Oe. As already stated, we believe that these samples are coarser grained and hence possess mostly VRM. The directions in Figure 13c are much more clustered than in Figures 13a or 13b, confirming the statistical result found earlier that undetermined polarities are associated with softer samples (Table 5).

Histogram plots of the data (Figures 14a and 14b) show that before structural correction the inclination and declination sets indicate a grouping of near-antipodal directions. This is most clear for directions from samples with the higher MDF. Gentle folding of the dike unit has contributed only a minor amount to the scatter of directions in Figure 13.

Pallister [1981] and Bailey [1981] have mapped the sheeted dike unit as coring the axis of an east-west trending, broad open syncline in the Ibra Valley. Most of our sites are located on either the north limb of this fold or in the axis. Structural corrections to most of the field directions in Figure 13 would be rather small. A few sites in the north (L40-L43, Figure 2) should be corrected for about a 15° south dip, while sites in the south (L3 and L57, Figure 2) could be corrected for about the same amount of northeast dip.

Areal Distribution of Polarities in the Ibra Valley

The reversed samples occur mixed with normal polarity samples at both the east and west ends

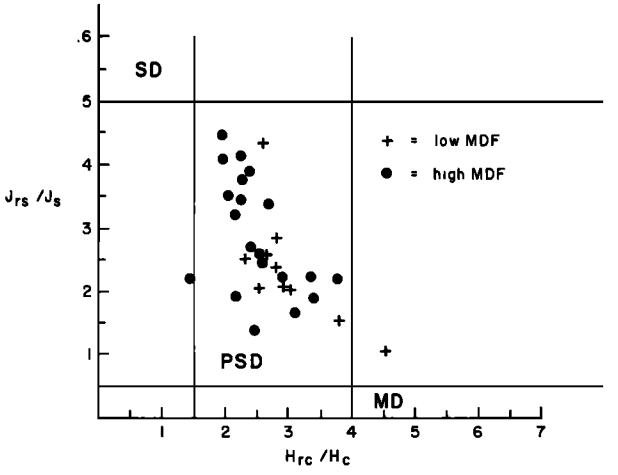


Fig. 11. Grain size domains defined by Day et al. [1977] according to the ratio of J_{RS}/J_S versus H_{RC}/H_C. The hysteresis values of samples listed in Table 2 are plotted and contrasted by MDF > 100 Oe. Most low MDF samples are larger in size than high MDF samples.

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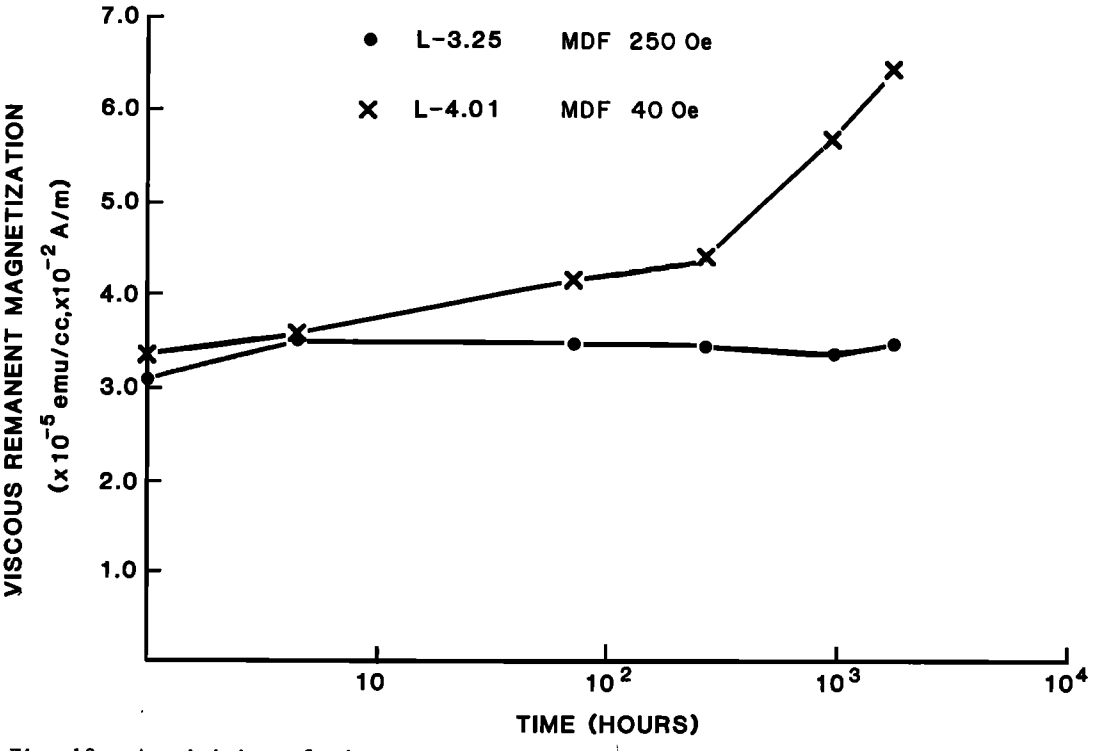


Fig. 12. Acquisition of viscous remanence in a 0.5-Oe field after demagnetization of the samples in 1000-Oe field.

of the map region (Figures 15 and 16). Sites with predominantly normal polarity samples form a 15-km-wide band toward the center of the outcrop area of the dike sequence (Figure 15). This band is bounded by site L36 in the east and site L57 in the west (Figure 2). In addition, all the polarities from the Wadi Kadir gabbro section (Figure 2) are normally magnetized [Luyendyk and Day, this issue]. The gabbro section is along strike from the normally magnetized dikes.

The degree of polarity mixing can be seen in a cross-strike plot of inclinations and declinations (Figure 16). At some sites normal and reversed directions are clumped (L3 at km 4), but at most other sites the directions are more dispersed (the directions for samples with MDF <100 Oe (7.96 x 10^3 A turn/m) are included in Figure 16). Other magnetic parameters including NRM and MDF show no discernible pattern with across-strike coordinate.

Both Figures 15 and 16 show that the area of largely normal polarity sites (km 9-16 approximately) has few or no intermediate directions (gray pattern of Figure 15), while areas with reversed directions contain most of the intermediate directions. If the occurrence of the U directions is correlated with the coincidence of R directions, the implication follows that these U directions may be due to normal polarity VRM overprinting an R direction or to combined N and R in an individual sample.

TABLE 4. MDF of NRM and TRM (0.75 Oe Applied Field)

Sample	Stable Declination		MDF of NRM, Oe	MDF of TRM, Oe
1.12	307°	N?	225	200
30.05	176°	R	40	145
32.04	2°	N	125	115
43.05	199°	R	40	270
64.04	351°	N	165	245
64.05	351°	N	240	250
64.06	357°	N	245	305

In our sampling program we were able to obtain multiple samples from 40 dikes. Of these, seven dikes clearly have both N and R samples (see Table 2). The dikes at the Al Ahmadi site (L3) are described in detail by Pallister [1981]. This outcrop displays a complete sequence of about 100 sheeted dikes with just over 100 m of stratigraphic thickness. We drilled two cores each from 12 individual dikes. Of these 12 dikes, five are normally magnetized, three are reversed, and four show both polarities [Laws, 1980]. The polarities are not spatially restricted but instead are intermixed. For the four dikes with dual polarities, two of them have a single chill margin (i.e., they are split), and a third is 1 m away from a dike with one chill margin. This opens the possibility that these dual polarity dikes were partially remagnetized by later intrusions which split them or close-by dikes (discussed by Pallister [1981]). Besides dike splitting by later intrusion, Pallister determined that 25% of the dikes intruded at the margins of older adjacent dikes. This event could also thermally remagnetize adjacent dikes. An observation which weakens the above idea of remagnetization by later intrusion is that the fourth dual polarity dike crosscuts all neighboring dikes and is therefore the latest event in the immediate proximity. An alternate possibi-

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TABLE 5. Magnetic Properties of Dike Rocks by Polarity

Variable		Number	Mean	Standard Deviation	Median	t_{nr}*, sig %	t_{nu}, sig %	t_{ru}, sig %
NRM, A/m	N	99	1.72	5.10	0.725	0.661	1.90	1.37
	R	45	2.50	7.19	0.50	48.9	93.8	82.5
	U	58	5.5	14.9	1.10			
MDF, Oe	N	92	227	170	188	0.130	3.23	2.05
	R	30	233	214	150	10.0	99.8	95.4
	U	44	144	126	108			
Susceptibility, SI	N	96	2.56 E-2	1.48 E-2	1.98 E-2	1.03	1.07	1.11
	R	38	2.22 E-2	1.60 E-2	2.04 E-2	69.5	71.0	72.6
	U	51	12.9 E-2	0.69	2.76 E-2			
Q_n	N	96	3.05	18.1	0.831	0.427	1.18	0.786
	R	38	4.25	13.0	0.630	33.0	76.0	56.6
	U	46	6.98	18.7	0.962			
H_C, Oe	N	13	251	122	225	0.816	1.38	2.49
	R	9	288	85	281	78.8	91.0	98.9
	U	13	193	90	197			
H_{RC}, Oe	N	14	599	165	568	0.965	2.50	1.52
	R	13	539	158	510	82.9	99.1	93.0
	U	17	446	175	380			
J_R, A/m	N	13	750	387	633	0.755	0.611	0.628
	R	9	638	307	530	77.0	72.6	73.1
	U	13	664	328	618			
J_S, A/m	N	13	3090	2020	2460	0.124	0.185	0.075
	R	8	3220	2490	2460	54.9	57.2	52.9
	U	8	3300	2800	2710			

*The means are significantly different at the given confidence level by the 't' test.

lity must be considered--that the polarity distribution in the outcrop is not only controlled by the sequence of dike intrusion but also by events probably associated with hydrothermal metamorphism. This phenomena has been noted in remagnetized flows from the Bermuda deep drill hole [Rice et al., 1980], and in N+R flows obtained in the Iceland deep drill hole [Bleil et al., 1982].

Virtual Geomagnetic Poles and Paleomagnetic Poles

Paleomagnetic poles were calculated for the sets of stable magnetization directions found for the dike samples. Average directions were found for each cooling unit, and these directions were averaged for each sampling site. The site directions were averaged to determine a paleomagnetic pole. We calculated poles for stable directions both corrected and uncorrected for local structure. The structural corrections were determined for each site according to the structural analysis described by Pallister [1981, Figure 2c]. The sites are distributed over three synclines and one anticline. Of the 28 sites used in the computation, 13 were corrected for south dip, seven for north dip, and eight are flat lying and were not corrected.

The paleomagnetic poles determined for corrected and uncorrected directions are not significantly different nor is one pole more precise than the other (Table 6). The same is true for poles calculated for the subset of 20 sites

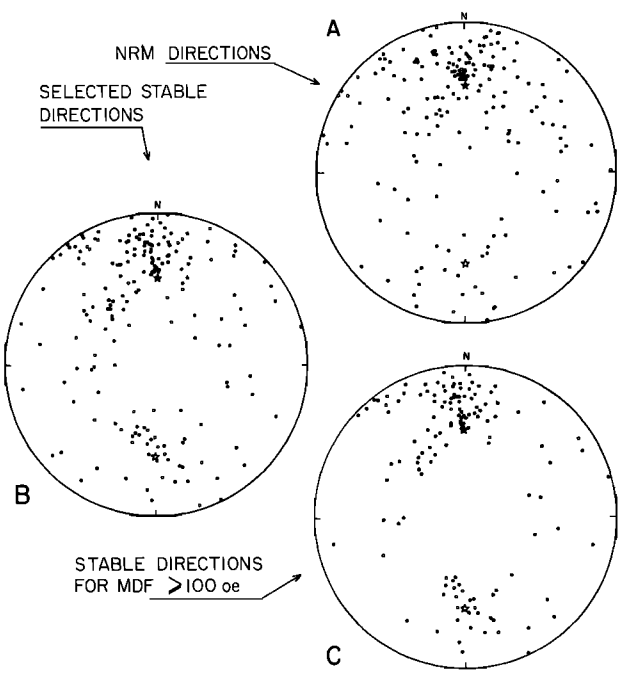


Fig. 13. Field directions of magnetization of the dike samples uncorrected for structure. (a) NRM directions. (b) Stable directions selected from AFDM series. (c) Those stable directions from samples with MDF > 100 Oe.

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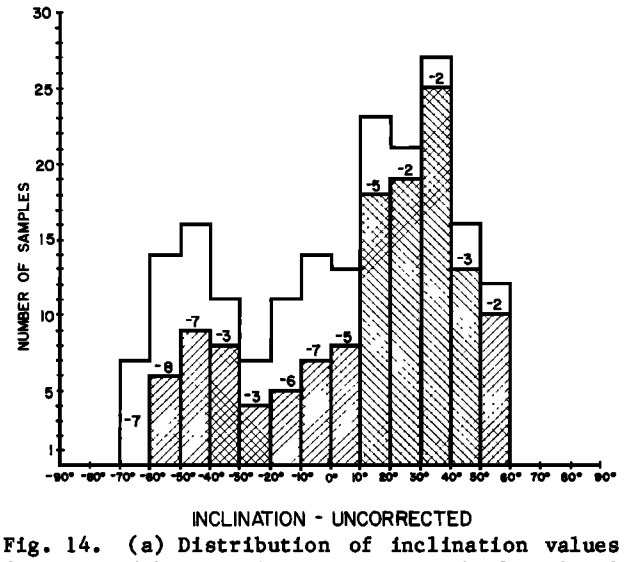


Fig. 14. (a) Distribution of inclination values for the dike samples, uncorrected for local structure. The open histogram is for all samples and the closed histogram is for samples with MDF > 100 Oe. The negative numbers indicate the difference in the number of total, low MDF samples for each class interval.

which required structural correction; the corrected versus uncorrected results are not significantly different nor is one more precise than the other.

We conclude from this analysis that the fold test cannot be used to discriminate postfolding magnetization. A pole calculated for a subset of 15 thermally demagnetized samples is not different at the 95% confidence level from the pole calculated from the AFDM sample set (Table 6, row 2). We also calculated poles separately for the normally and reversely magnetized sample set. The resulting poles are not different from the poles in Table 6, rows 1 and 2 or each other, and therefore the reversal test is passed (N is 77.4N, 271.8E, $\alpha 95 = 6.7$; R is 69.0N, 241.9E, $\alpha 95 = 17.8$).

The location of the Table 6, row 2 pole is different from the geocentric axial dipole (Figure 17). The Samail ophiolite dike sequence pole is presently far sided, suggesting northward movement of the ophiolite since magnetization. However, the pole is not as far sided as the Cretaceous pole for Africa (Figure 17) [Gough et al., 1964]. In fact, relative to the Africa pole, the dikes pole is near sided. These results can be interpreted as north-south closure of the ophiolite relative to Arabia-Africa (after magnetization), obduction and emplacement of the ophiolite onto Arabia, then northward translation of Arabia (plus ophiolite) and Africa. This scenario is similar to one discussed by Coleman [1981]. The amount of relative closure (ophiolite southward relative to Arabia, or Arabia northward relative to the ophiolite) is 13.1° ± 11.0° (Table 6, row 6; see Beck [1980] for calculation method).

Discussion and Conclusions

Probably the most significant observation reported here is that of apparent opposite magnetic polarities. Some important questions to ask now are whether the reversals are primary (recorded according to the hypothesis of Vine and Matthews [1963] and if the reversals can be correlated with known magnetic polarity stratigraphy, or whether either or both polarities are due to a secondary magnetization.

Models for secondary magnetization of the dikes must take into account the probably thermal history of the dikes at a spreading center. Our dike samples contain magnetite, hematite, ilmenite, and maghemite. The magnetite and hematite probably formed in the ocean crust. The maghemite is probably a result of subaerial weathering [see Beske-Diehl and Banerjee, 1979]. Magnetite is commonly seen in the dike rocks of ophiolites [Levi and Banerjee, 1977; Banerjee, 1980]. The magnetite in the Samail dikes could be exsolved from titanomagnetite by deuteric oxidation at the spreading center [Banerjee, 1980]. Thus the primary magnetization in the dikes is a high-temperature CRM or TRM. In addition, the pervasive hydrous metamorphism present in the dike sequence [Gregory and Taylor, 1981; Pallister, 1981] points to the possibility of secondary magnetization due to the formation of hematite from the oxidation of magnetite and ilmenite at a later time. We propose that N polarities are held in the exsolved magnetite and that the R polarities are associated with later secondary hematite. Some of the U samples possess unseparated portions of both primary N and secondary R directions. The observation that both N and R (and also U) polarities are found in the same cooling unit is satisfactorily explained

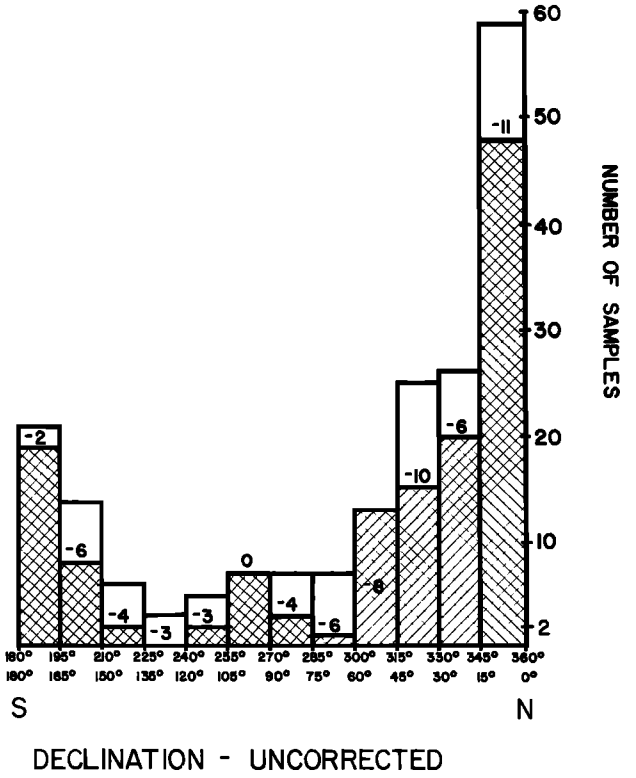


Fig. 14. (b) Distribution of uncorrected declinations.

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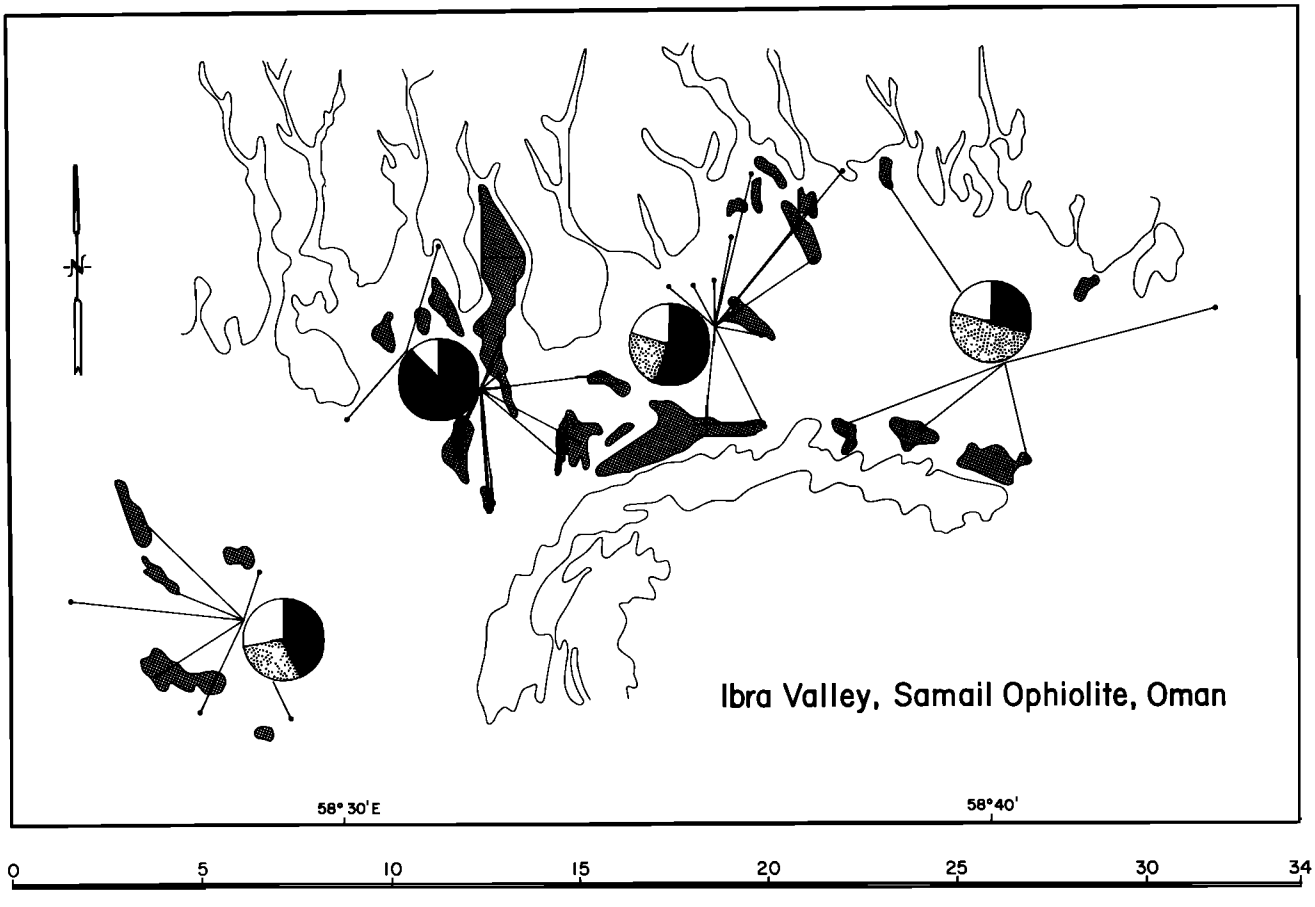


Fig. 15. Pie diagrams of the areal distribution of polarities for groups of sites (dots); black is normal polarization, white reversed, and dot patterns are intermediate directions. Sites are identified in Figure 2. Samples from the canyon north of the most normal pie (Wadi Kadir) are also normal but are not shown here [Luyendyk and Day, this issue].

by secondary magnetization due to hydrothermal metamorphism.

Several lines of evidence support the idea that the original polarity of the dikes is normal and that the secondary component is reversed. One fact is the areal and numerical predominance of N polarity compared to R. Besides observing more N than R polarities in the dike sequence, the lava flows and gabbros in the study area are also normally magnetized. A second observation is that the TDM experiments suggest that the R magnetization apparently resides in hematite. The TDM experiments on N samples show that this polarity is in lower blocking temperature grains which are magnetite.

Another mechanism to explain the observed opposite polarities is self-reversal in titanomagnetites. The possibility of self-reversal is intricate and may seem remote, but it nevertheless merits a careful evaluation. Instances of reproducible self-reversal above room temperature have been reported in cation deficient titanomagnetites in an oceanic basalt by Ryall and Hall [1975] and in an alkali basalt by Schult [1976]. In the cases reviewed, the self-reversal mechanism probably involves moderate temperature exsolution of a Ti-poor daughter phase from a host titanomagnetite which produces a reversed interaction field. The daughter phase retains the original polarity, while its interaction field reverses the polarity of the host. Because of their differing Ti content, the host and daughter phase yield differing Curie temperatures in the same sample. Also, the two directions are blocked at different temperatures. Even though this process does not appear compatible with the thermal history of the Samail sheeted dike sequence, six samples were tested for reproducible self-reversal. Samples of both normal and reversed polarities and with a range of MDF values were included in the experiment. The experiment was conducted by imparting a total TRM in a field of 0.75 Oe to the samples. All specimens acquired a remanence parallel to the applied field, and subsequent AFDM was not indicative of any reversed component.

Thus these rocks are not now self-reversing; however, the possibility of nonreproducible self-reversal still exists [Merrill, 1975]. This phenomena has been studied in diorites by Merrill and Grommé [1969]. It was found that magnetite was responsible for a normal polarity component and titanohematite (10 to 15 mol % ilmenite in hematite) carried a self-reversed component [Merrill and Grommé, 1969]. This relationship was clearly illustrated by thermal demagnetiza-

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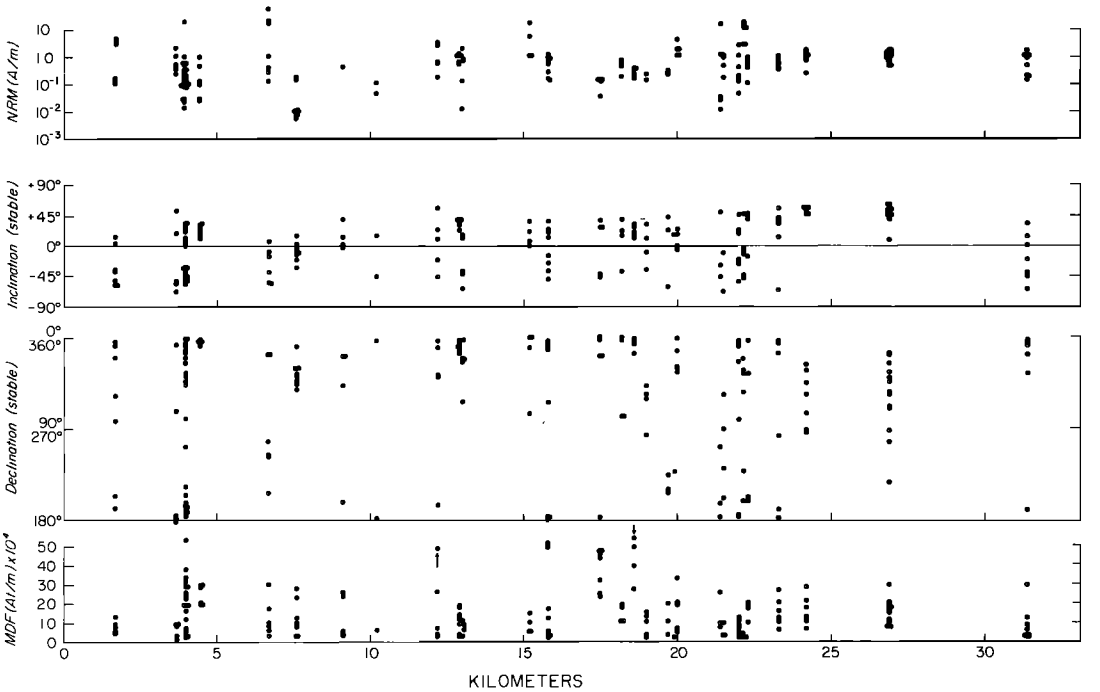


Fig. 16. Magnetic parameters versus the cross-strike coordinate (origin at the left-hand side of Figure 15). All the data are shown; i.e., they are not sorted according to MDF. The directions are not corrected for structure.

tion experiments carried out above the Curie temperature of magnetite (580°C). Although there are similarities in magnetic mineralogy, the nonreproducible self-reversal mechanism of Merrill and Grommé [1969] has several requirements not met by the samples from the Samail sheeted dike sequence. In their case the titanohematite and magnetite grains generally occurred independently in crystals of contrasting morphology; this was not observed in our samples. In addition, we did not detect titanohematite in a microprobe experiment. Magnetite, hematite, and ilmenite within sample L3.01 were probed. No Ti was detected in the host magnetite or within the hematite. No Fe was detected within the ilmenite lamellae. Also important is the time requirement of the self-reversing mechanism. Merrill and Grommé [1969] point out that this would require an extremely long cooling history, such as that associated with a batholith. The cooling rate of a submarine dike is too short a time period to yield a nonreproducible self-reversal of this type. Finally, if, as in the samples of Merrill and Grommé [1969], the hematite in the samples of this study was self-reversing, then the thermal demagnetization studies should have shown some difference in field directions at temperatures above 580°C. From these facts, we believe that self-reversal in the Samail dikes is unlikely; however, without

TABLE 6. Paleomagnetic Poles for the Dike Sequence (28 Sites, 63 Units, 99 Samples)

	Im	Dm	Lat	Long	k	α95	α63	R	dp	dm
Structurally uncorrected[a]	18.2	354.6	75.6N	260.5E	6.7	11.4	6.5	23.9	6.2	11.9
Structurally corrected[b]	18.3	355.7	76.0N	256.5E	5.9	12.3	7.0	23.4	6.7	12.8
TDM uncorrected (5 sites 15 samples)	38.0	351.6	82.1N	320.0E	10.9	24.2	12.1		16.9	28.6
TDM corrected	29.1	1.0	82.8N	230.8E	5.8	18.0	9.9		11.0	19.9
Reference pole[c]			62.0N	259.0E	33.9	3.0			4.0	4.0

Rotation and translation of site relative to reference pole (row 2 versus row 5)

[a]Paleolatitude, 9.3°N; A95 = 10.7.
[b]Paleolatitude, 9.4°N; A95 = 10.6.
[c]African = African Lupata volcanics 106-110 m.y. [Gough et al., 1964] (A95 = 4.0).
Rotation = 5.1° CW ± 13.3°. Inclination anomaly = 25.6° ± 13.7° (steeper). Translation = 13.1° southward ± 11.0°.

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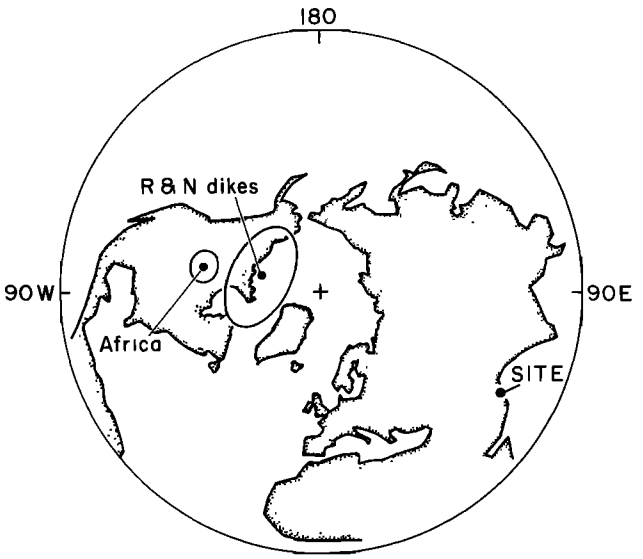


Fig. 17. Paleomagnetic pole for the Samail ophiolite dike sequence near Ibra. The pole is calculated for structurally corrected site data for samples with MDF > 100 Oe (Table 6, row 2). Also shown is the Cretaceous Africa pole of Gough et al. [1964].

more exhaustive experiments we cannot dismiss the possibility entirely.

Demagnetization treatments apparently remove the VRM components quite successfully. However, both the primary normally magnetized and secondary reversely magnetized grains have overlapping coercive force spectra, and in some cases the two directions intermix to produce incoherent U directions. This same problem was encountered by Rice et al. [1980] but to a greater degree. In many cases, though, we are able to separate out R directions which reside in higher coercivities of multicomponent direction samples, and, of course, many samples have only one discernible directional component (Figures 5, 6, and 8). We feel safe in concluding that truly original stable TRM and CRM directions have been defined in our study.

Presuming that the R polarities were recorded near the spreading center, the question is now whether these reversed polarities can be correlated with the known geomagnetic time scale. The radiometric age of the ophiolite at Ibra (94-97 m.y.) places it in the Cenomanian according to either the London [Geological Society of London, 1964] or van Hinte [1976] time scales. However, van Hinte [1976] points out the controversial nature of the age assigned for the Albian-Cenomanian boundary. Individual estimates range from 93 to 110 m.y. before present. Lanphere and Jones [1978] have placed the Cenomanian-Albian boundary at 96 m.y., using K-Ar mineral ages. This would place the ophiolite age here as late Albian and early Cenomanian. During most of the Late Cretaceous the magnetic field was of normal polarity [Lowrie et al., 1980]. The marine magnetic anomaly record suggests that the latest reversal to precede the normal interval was in the Aptian (anomaly M0 of Larson and Hilde [1975]. Magnetic stratigraphy studies indicate that several short reversals may have followed M0. The results of these studies have been synthesized by Lowrie et al. [1980], who noted that six or more short reversed events may occupy a period after M0 from the late Aptian to the middle Cenomanian. Keating et al. [1975] reported an Albian-Cenomanian reversal, which is the redated Akoh reversal of Sasajima and Shimada [1976]. Keating et al. [1975] report a redating to 92-104 m.y. B.P. for this formation, but since they did not document this date, we refrain from attempting any correlation. Three reversals were noted in the Albian section of Deep Sea Drilling Project (DSDP) site 263 by Green and Brecher [1974]. Keating and Helsely [1978] report that four Albian reversals have been seen in DSDP sites. We have previously found a short reversed polarity section in the middle Albian clays of DSDP site 258. (Of 99 samples from 184 m of section, 19 samples in a 7-m-long interval from cores 24 and 25 are reversed; data are available on request.) In an intensive study of the Cretaceous Umbrian limestones and marls of Italy, Lowrie et al. [1980] and VandenBerg et al. [1978] found a short reversed interval in the late Aptian of the Valdorbia section (ISEA event). However, it could not be found in nearby equivalent sections. The Valdorbia reversal is younger than anomaly M0 [Lowrie et al., 1980] but older than the Ibra reversal(s). VandenBerg and Wonders [1980] noted two late Albian reversals and three Cenomanian reversals in stratigraphic studies of the southern Alps (VM and QS events). None of the reported short events appears consistently throughout the various stratigraphic studies yet undertaken. Nevertheless, existing data suggest that short reversed events follow the M0 reversal. However, high sedimentation or spreading rates are necessary to resolve them. In this regard, the high spreading rate proposed for the Samail ophiolite is significant [Tilton et al., 1981]. Therefore, the reversal(s) we have observed here may correspond to a poorly known mixed polarity interval, slightly younger than the M0 anomaly, which is imperfectly preserved in the geologic record.

We come now to the central question of our work: Can the oceanic sheeted dike layer contribute significantly to the marine magnetic anomaly record? We believe probably not. Beneath the ocean floor, we would expect that the Samail dike layer possessed NRM magnetic intensity values similar to Table 1 and NRM directions like those shown in Figure 13a. Although the NRM magnitude and Q factors are reasonably high, the NRM directions are widely dispersed except for a cluster near the earth's dipole field direction. This dispersion significantly decreases the likelihood of a net volume vector magnetization in the dike layer, strong enough to make a significant contribution to a magnetic anomaly. Even if the directions were all coherent, the NRM intensities indicate that the dike layer would contribute only 30 to 60 nT to a magnetic anomaly measured 5 km away at the ocean surface. A possible contributor to the dispersion of directions is a self-reversal effect in secondary titanohematite. If we have observed this effect here, then there is reason to believe it can occur generally and

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thus confuse any inherent direction coherence in dike units of the ocean crust.

The dispersion of directions, or the U = undetermined polarity found here, is due to overprint of normal polarity VRM on R directions and to incomplete overprint of secondary R directions on primary N directions. Completely counter to our argument here, Rice et al. [1980] suggested that secondary magnetization due to greenschist metamorphism could very well homogeneously remagnetize the entire dike layer and thereby decrease its dispersion and increase its contributing strength. The fact remains though, in both the Samail dikes and Bermuda case, that the secondary magnetization increased dispersion. Their argument is good in principle, but its truth rests with the degree and extent of metamorphism and remagnetization.

Finally, we want to point out that within error limits, the dikes are magnetized exactly along their average strike. Pallister [1981] determined this strike to be between 347° and 353° ± 15° to 25°; the average paleomagnetic declination is 356° ± 12°. This means that the ancient spreading was oriented about north-south. The insignificant rotation anomaly indicates that the ophiolite was obducted with no net rotation. The paleolatitude of the unit (Table 6) indicates that the ophiolite was located in the Tethyan Ocean just north of the equator (9°N).

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